

Supplementary Material: Graph-Guided Mixture-of-Experts for Unified Multimodal Mental Health Assessment

Anonymous submission

Experimental Pipeline Overview

The central empirical question of the main paper is the graph-routing investigation. To isolate its effect, we compare five graph-routing ablation variants (V0–V4) against non-graph baselines. When initial results showed no improvement on DAIC/MOSEI and degradation on FI personality, we deployed a homophily diagnostic comparing real graph edges against random pairing, alongside targeted fixes including a K -sensitivity sweep ($K = 5, 10, 15, 20$) and learning a per-task λ weight instead of fixing it at 0.5. These diagnostics confirmed that graph routing does not help in this setting because the required signal is either absent or unused.

Leakage Protocol Details

Graph Construction per Split

For each dataset, we construct KNN graphs using one of three leakage-safe modes (see Section 3, “Method,” of the main paper):

- **Inductive:** the graph is built on training-split embeddings only. Validation and test nodes connect exclusively to their K nearest *training* neighbours; they never connect to each other.
- **Split-local:** separate graphs are built independently for train, validation, and test. No edge crosses a split boundary.
- **Transductive** (ablation only, not leakage-safe): a single graph is built over all splits jointly, so validation/test nodes may connect to each other. Retained only to quantify how much a naive (non-leakage-safe) protocol would inflate results.

DAIC-WOZ: 189 sessions (107 train / 35 val / 47 test), subject-independent — no participant ID appears in more than one split. CMU-MOSEI and ChaLearn FI use their official utterance-/clip-level splits (see Section 3, “Method,” of the main paper). All splits are validated by an automated leakage audit (9/9 checks passing: inductive train/test, split-local val/test, transductive cross-split-edges-expected, bug-injection detection, and LLM-feature independence).

Table 1: Complete hyperparameter configuration.

Parameter	Value
Common dimension d	256
MMoEEx experts M	8
Expert hidden size (both configurations)	256
GraphSAGE layers	2
GraphSAGE hidden size	126
GraphSAGE aggregation	Mean
GAT heads / head dim (ablation)	3 / 42
KNN neighbours K	10 or 15 (per variant)
Edge similarity metric	Cosine
Graph routing weight λ	0.5 (fixed, not tuned)
Learning rate	3×10^{-4}
Optimizer	AdamW
Batch size	32
Epochs (max)	150
Early stopping patience	20
Projector freeze epochs	20
Temperature balance T	3.0

Hyperparameters

Complete Ablation Results

Row 4 (simple KNN label voting, no learned GraphSAGE router at all) reaches the best DAIC AUROC in the entire ladder. The learned router (rows 5–6) does not improve on this — the clearest single illustration that the learned component underperforms even the simplest graph-based heuristic on the primary clinical task.

Graph-Sensitivity Sweep

Learned Routing Weight

Additional Figures

Expert Routing Analysis

Figure 6 shows the real MMoEEx routing-weight distribution (non-graph baseline): each task concentrates weight on a small subset of its assigned experts rather than routing uniformly, indicating the gates learn task-differentiated behaviour even where the aggregate metric for that task declines relative to a unimodal baseline (see main paper).

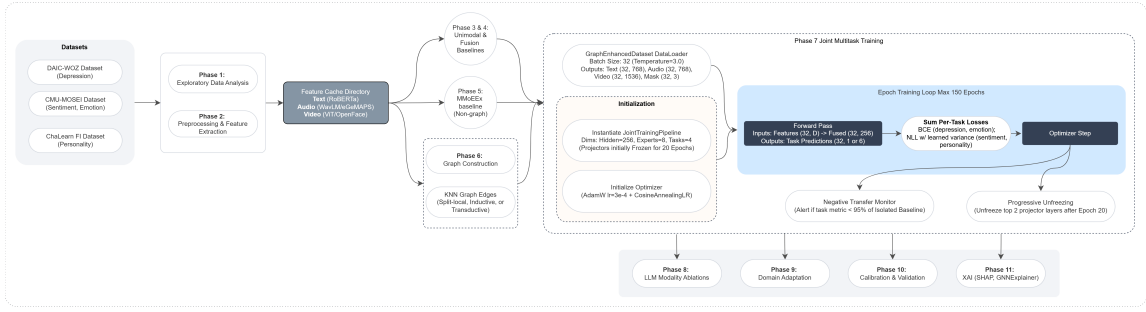


Figure 1: Implementation detail of Phases 5–11: the non-graph MMoEEEx baseline (Phase 5) and the graph-routed joint multitask training (Phase 7) share the same DataLoader, initialization, and epoch training loop, differing only in whether the graph router contributes to expert routing; downstream phases (LLM ablations, domain adaptation, calibration, explainability) follow.

Table 2: Full ablation ladder, real labels and real evaluation throughout. Each row adds one component cumulatively. Source: artifacts/tables/ggmoe_results.csv, mmoe_ex_results.csv, unimodal_baselines.csv, knn_voting_results.csv.

		DAIC AUROC	MOSEI Sent. CCC	MOSEI Emotion AUROC	FI Avg CCC
0	Trivial (majority class)	0.500	0.000	0.500	0.000
1	+ Unimodal (best modality)	0.535	0.512	0.771	0.458
2	+ Gated late fusion	0.496	0.623	0.771	0.000
3	+ MMoEEEx (no graph)	0.493	0.493	0.722	0.578
4	+ KNN voting (no learned router)	0.630	0.443	0.676	0.390
5	+ Graph router (V0, inductive $K=10$)	0.515	0.517	0.719	0.542
6	+ Graph router (V3, inductive $K=15$)	0.525	0.510	0.734	0.544

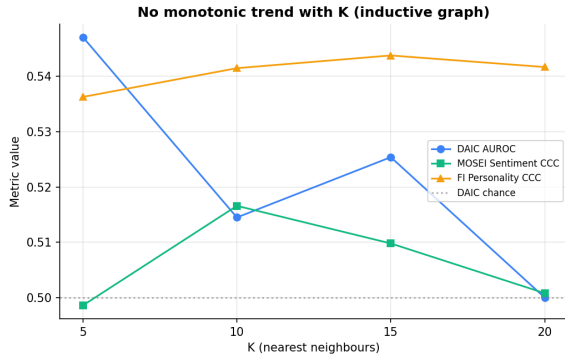


Figure 2: No monotonic trend with $K \in \{5, 10, 15, 20\}$ (inductive graph). $K=20$ lands DAIC AUROC exactly at chance (0.500).

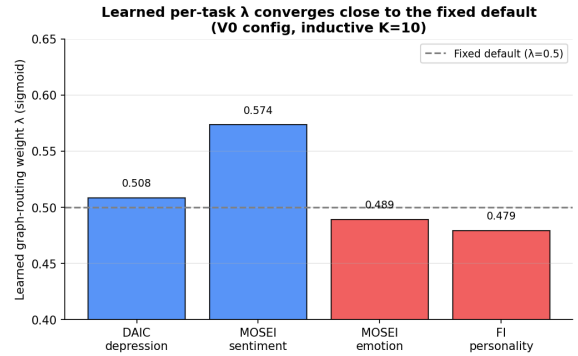


Figure 3: Learned per-task λ (V0 configuration) vs. the fixed default (0.5). Moves down for FI/MOSEI-emotion and up for MOSEI-sentiment/DAIC, consistent with the homophily diagnostic (main paper, Table 2), but only slightly.

Explainability Case Studies

Beyond the routing-performance question, the main paper’s Introduction motivates two further, separately-evaluated hypotheses: that mental health assessment can be made more transparent by characterizing depression jointly with sentiment, emotion, and personality rather than in isolation, and that a graph over similar individuals can give both a visual and a machine-readable account of what informed a prediction. This section reports both.

Multi-Construct Characterization

For 9 case studies (3 per dataset), we pass the same sample’s shared representation through all four task heads — not only the head matching its own source dataset — producing a depression probability, sentiment score, six emotion probabilities, and five personality-trait scores for the same person in one forward pass. On a representative positive-sentiment MOSEI case, the profile is internally coherent: happiness probability 0.89, sentiment score +0.62, with sadness/anger/fear all below 0.09 — a plausible

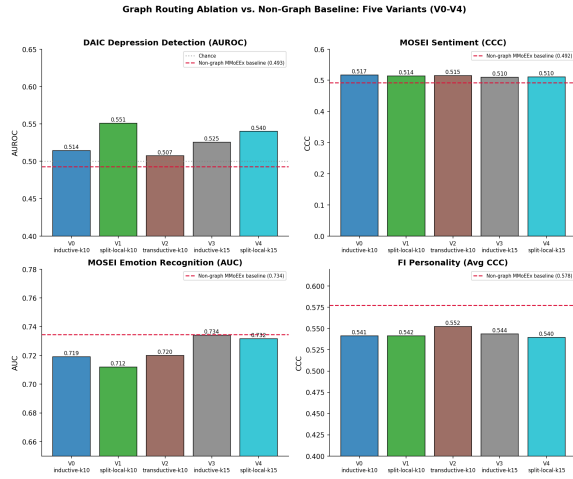


Figure 4: Graph routing ablation vs. non-graph MMoEEEx baseline (red dashed line), all five variants across all four tasks.

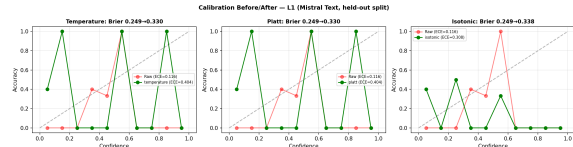


Figure 5: Calibration reliability diagrams (temperature/Platt/isotonic scaling vs. raw) for the L1 (Mistral text) DAIC depression classifier, held-out split. Monotonic recalibration changes reliability (ECE/Brier) but not ranking-based discrimination (AUROC), since AUROC is invariant to monotonic transforms of the score.

joint characterization rather than four unrelated numbers. Related work explicitly separating personality-trait context from acute depression symptoms in an interpretable graph model (Vyalla et al. 2026) supports this direction; each profile is stored as a `multi_construct_profile` field in `artifacts/tables/phases11_xai_results.json`.

Graph-Based Explanation

For each case study we generate a visual record (SHAP per-modality beeswarm plot, GNNExplainer subgraph diagram, and a GraphXAIN-style (Cedro and Martens 2024) natural-language narrative panel, Figure 7) and a machine-readable record (SHAP values, top- k neighbour identities and cosine distances, expert routing weights, the multi-construct profile, and counterfactual/perturbation deltas). The narrative is generated by prompting an LLM (Mistral-7B) with the SHAP values, subgraph structure, and sample metadata, used purely as *post-hoc* explanation of an independently-computed prediction — not as reasoning inserted into the prediction pipeline itself. This distinction matters: chain-of-thought-style reasoning improves mental-health text classification when used to produce the prediction (Patil and Gedhu 2025), but can be unfaithful or reduce reliability when embedded in

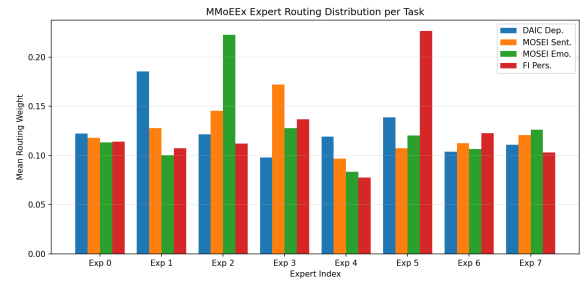


Figure 6: Mean MMoEEEx routing weight per expert and task (non-graph baseline). Rows/legend are tasks; columns are the 8 experts.

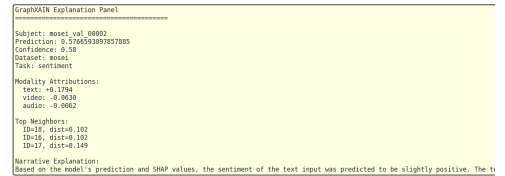


Figure 7: GraphXAIN-style narrative panel for a representative MOSEI case, combining SHAP attribution, sub-graph structure, and sample metadata into a post-hoc natural-language explanation.

the clinical decision itself (Wu et al. 2025).

Every SHAP, perturbation, and counterfactual value must be computed under each sample's own dataset-native routing (main paper, Section 3): DAIC is evaluated text-only and FI video-only, so a modality outside a dataset's native routing shows *exactly* zero attribution, not a small residual. Across 90 perturbation tests, only 47 are non-zero, exactly matching each sample's routing; the remaining 43 are exactly zero rather than merely small. This is a stronger faithfulness check than a ranking comparison: it directly confirms the explanation pipeline reflects each dataset's real computation.

The explanation graph is built on the model's own trained fused representation rather than raw concatenated features, so reported neighbours are the ones the router and task heads actually operate on. This does not, however, improve DAIC label homophily: on the same 35 DAIC validation samples used for the main paper's homophily diagnostic, label agreement is 0.577 for raw features, 0.537 for random pairing, and 0.463 (below random) for the trained representation, since task-supervised training optimizes for the final decision boundary rather than for preserving neighbourhood structure. DAIC subgraph explanations should be read as faithful to the model's own computation, without a validated claim that the shown neighbours are similar in a depression-relevant sense; MOSEI and FI do not share this limitation, consistent with the main paper's homophily diagnostic finding real label

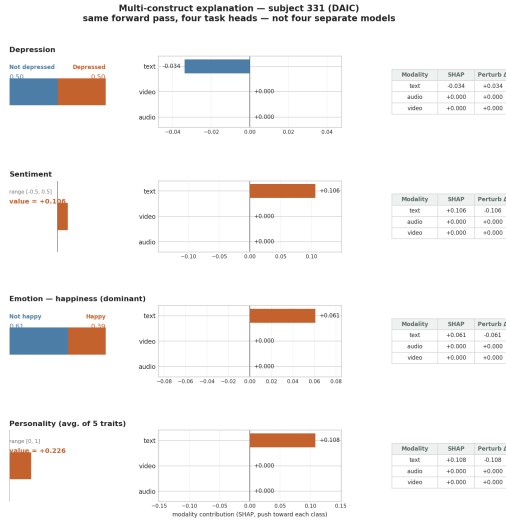


Figure 8: Multi-construct force plot: one DAIC subject explained through all four task heads in a single figure. Audio and video read exactly zero on every row, visualizing DAIC’s text-only routing directly rather than leaving it implicit.

signal for those tasks.

References

- Cedro, M.; and Martens, D. 2024. GraphXAIN: Narratives to Explain Graph Neural Networks. *arXiv:2411.02540v3*. Used for: GraphXAIN narrative explanations, LLMs for GNN explainability.
- Patil, A.; and Gedhu, A. K. 2025. Cognitive-Mental-LLM: Evaluating Reasoning in Large Language Models for Mental Health Prediction via Online Text. *arXiv preprint arXiv:2503.10095*. Used for: Chain-of-Thought / Self-Consistency / Tree-of-Thought reasoning improving mental-health text classification, motivating LLM-based narrative explanation.
- Vyalla, R. R.; Prasad, K.; Anand, A.; Cambria, E.; Ji, S.; Alamri, F. S.; and Wang, Z. 2026. Psychologically-Grounded Graph Modeling for Interpretable Depression Detection. *arXiv preprint arXiv:2604.24126*. Used for: directly relevant precedent distinguishing personality trait context from acute depression symptoms via psychologically-grounded, interpretable graph modeling.
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